Chit 1 — SJF Preemptive

Input:
3
0 5
1 3
2 6
Short Output:
Gantt → P1,P2,P1,P3
Avg WT = 3.00, Avg TAT = 7.67
Explanation: Chooses process with the smallest remaining time at every moment (preemptive SJF).

Chit 2 — Round Robin (Arrival, q=2)

Input:
3 2
0 5
1 4
2 2
Short Output:
Gantt → P1,P2,P3,P1,P2,P1
Avg WT = 4.33, Avg TAT = 8.00
Explanation: Each ready process gets a time quantum q; preempt and cycle to ensure fairness.

Chit 3 — Disk Scheduling SSTF

Input:
5 50
55 58 39 18 90
Short Output:
Order → 50 → 55 → 58 → 39 → 18 → 90
Total head movement = 120
Explanation: Shortest Seek Time First: always serve the nearest pending cylinder.

Chit 4 — Round Robin (same as Chit 2)

Input:
3 2
0 5
1 4
2 2
Short Output:
Gantt → P1,P2,P3,P1,P2,P1
Avg WT = 4.33, Avg TAT = 8.00
Explanation: Same RR scheduling with arrivals and quantum 2.

Chit 5 — Producer–Consumer (Semaphores + Mutex)

Input:
(no input)
Short Output:
Sample lines → Produced 1 … Consumed 1 … Produced 2 … Consumed 2 …
Explanation: Bounded buffer: producer inserts; consumer removes; semaphores count slots/items; mutex protects critical section.

Chit 6 — Banker's Algorithm (Safety)

Input:
3 3
Max:
7 5 3
3 2 2
9 0 2
Alloc:
0 1 0
2 0 0
3 0 2
Avail:

3 3 2
Short Output:
Unsafe state
Explanation: Computes Need = Max – Alloc and checks if all processes can finish in some order (safe sequence).

Chit 7 — IPC via System V Shared Memory

Input:
(no input)
Short Output:
Sample lines → Client read: Hello from server … ; Server sees: ACK from client
Explanation: Parent and child share a memory segment for fast data exchange; no kernel copying after attach.

Chit 8 — LRU Page Replacement

Input:
3 12
7 0 1 2 0 3 0 4 2 3 0 3
Short Output:
LRU faults = 9
Explanation: Evict the page that has not been used for the longest time in the past.

Chit 9 — Optimal Page Replacement

Input:
3 12
7 0 1 2 0 3 0 4 2 3 0 3
Short Output:
Optimal faults = 7
Explanation: Evict the page whose next use is farthest in the future (ideal baseline).

Chit 10 — Compare FCFS/LRU/Optimal (Faults)

Input:
3 12
7 0 1 2 0 3 0 4 2 3 0 3
Short Output:
FCFS = 10, LRU = 9, Optimal = 7
Explanation: Compares miss counts across three policies on the same reference string.

Chit 11 — Producer–Consumer (threads)

Input:
(no input)
Short Output:
Sample lines → Produced k / Consumed k (buffer index rotates)
Explanation: Same as Chit 5; emphasizes POSIX threads + counting semaphores.

Chit 12 — Readers–Writers (1st readers' preference)

Input:
(no input)
Short Output:
Sample lines → Reader i reads v … Writer j writes v+1 …
Explanation: Multiple readers can read concurrently; writers get exclusive access; first reader locks writers, last reader unlocks.

Chit 13 — SCAN (Elevator)

Input:
8 50 200
176 79 34 60 92 11 41 114
Short Output:
Order → 50 → 60 → 79 → 92 → 114 → 176 → 41 → 34 → 11
Total head movement = 291
Explanation: Moves in one direction servicing requests, then reverses, reducing variance vs FCFS.

Chit 14 — Banker's (multi)

Input:
Same as Chit 6
Short Output:
Unsafe state
Explanation: Same safety check on the given matrices, multi-process/multi-resource.

Chit 15 — Round Robin Gantt (q=2)

Input:
Same as Chit 2
Short Output:
Gantt → P1,P2,P3,P1,P2,P1
Avg WT = 4.33, Avg TAT = 8.00
Explanation: Shows time-sliced execution order (visual Gantt) with averages.

Chit 16 — SJF Preemptive (with arrivals)

Input:
Same as Chit 1
Short Output:
Gantt → P1,P2,P1,P3
Avg WT = 3.00, Avg TAT = 7.67
Explanation: Shortest remaining time next; preempts when a shorter job arrives.

Chit 17 — Page Replacement (Compare)

Input:
Same as Chit 8
Short Output:
FCFS = 10, LRU = 9, Optimal = 7
Explanation: Quick side-by-side of miss counts to discuss policy trade-offs.

Chit 18 — Disk: SSTF / SCAN / C-LOOK

Input:
Same as Chit 13
Short Output:
SSTF total = 120, SCAN total = 291, C-LOOK total = 321
Explanation: Shows how different head movement strategies affect total seek distance.

Chit 19 — Readers–Writers (threads)

Input:
(no input)
Short Output:
Sample lines → Reader… Reader… (parallel) then Writer… (exclusive)
Explanation: Same concept as Chit 12; emphasizes thread joins and clean teardown.

Chit 20 — Producer–Consumer (counting semaphores)

Input:
(no input)
Short Output:
Sample lines → Produced 1,2,… and matched Consumed 1,2,…
Explanation: Counting semaphores track empty/full slots; mutex ensures mutual exclusion.